Part C in one page — market making on hftbacktest: what made money, what didn't

The question behind all of Part C: can reinforcement learning improve a published market-making strategy, measured on days it never saw? This page is the summary; each phase has its own writeup, linked in §4.

> **At a glance**
>
> - **Best held-out result:** OBI on XRPUSDC with a markout-driven spread, **+4.91%** over eleven held-out sessions, **+1.38% without the crash day**, positive on **9 of 11** sessions at ~500 fills a day.
> - **Most robust held-out result:** OBI on SUIUSDC (C8, seven held-out sessions on a 24/7 split), **+2.52%** at 175 fills a day, and still positive without the crash (+1.77%), without its best session (+1.52%), without the maker rebate (+2.05%), with a 1 bp maker fee (+1.13%) and at 4× order latency (+1.30%). XRP's OBI gain, by the same stress tests, is the crash session.
> - **The fair price matters more than anything else tried — if it were known.** Quoting around the mid 1–3 s ahead (C9) turns ETH GLFT from −3.25% to +4.30–19.67% held out; but the best forecast of the mid explains 2–10% of its 1 s move, where about 40% would be needed, and no forecast, large-move rule or DAgger student captures the gain.
> - **Everything on ETHUSDC still loses held out**, bar the rebate. The best, the same rule on OBI, is at −0.62%, against −6.40% for the widened baseline and −7.14% for the published parameters.
> - **The two things that helped were both hand-written rules:** a wider, per-asset spread floor (C6) and widening after toxic fills (C7).
> - **Learning did not beat them.** TD3 on C6's widened strategies (two recipes, five seeds), GRPO- and PPO-style parameter search, extra order-book and BTCUSDT inputs, fill targets, a reward that discounts inventory PnL and networks up to 170 times larger all lose to the best fixed rule held out; so does RL on top of that rule (C8). DAgger with a teacher that sees the future mid (C9) halves ETH GLFT's loss without making it pay. A circuit breaker helps the losing strategies on ETH; extra controls chosen on training data hurt OBI.
> - **Why learning keeps losing** (§6): reward-driven RL learns from a signal buried in mark-to-market noise, and every method — RL or imitation — is capped by what the book tells it, which C9 measured at R^2 0.02–0.10 of the next second's move. The next step is new inputs from outside the book (§7), most of them free.

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1. Setup, in five lines

- **Data:** Binance USDⓈ-M perpetual L2 books and trades, 2025-08-26 to 2025-09-25, replayed tick by tick in hftbacktest with queue position and synthesised order latency.
- **Split:** everything is chosen on sessions before 2025-09-15 and reported on the eleven sessions from 09-15 to 09-25 (C1–C7, C4). C8 re-splits the same month into 24 training and 7 held-out sessions (09-19 to 09-25). Nothing is chosen on held-out data.
- **Deep assets:** ETHUSDC and XRPUSDC, 11 held-out sessions each. Seven more assets with 3 held-out sessions each.
- **Strategies:** the five market makers hftbacktest publishes — GLFT, a volatility grid (`hfgrid`), order-book imbalance (`obi`), and two pair strategies (`apt`, `basis`) on XRP.
- **Score:** cumulative held-out return in % of a 100,000 book; fills and per-session consistency beside it, so "stopped trading" never passes for "made money".

2. The track record

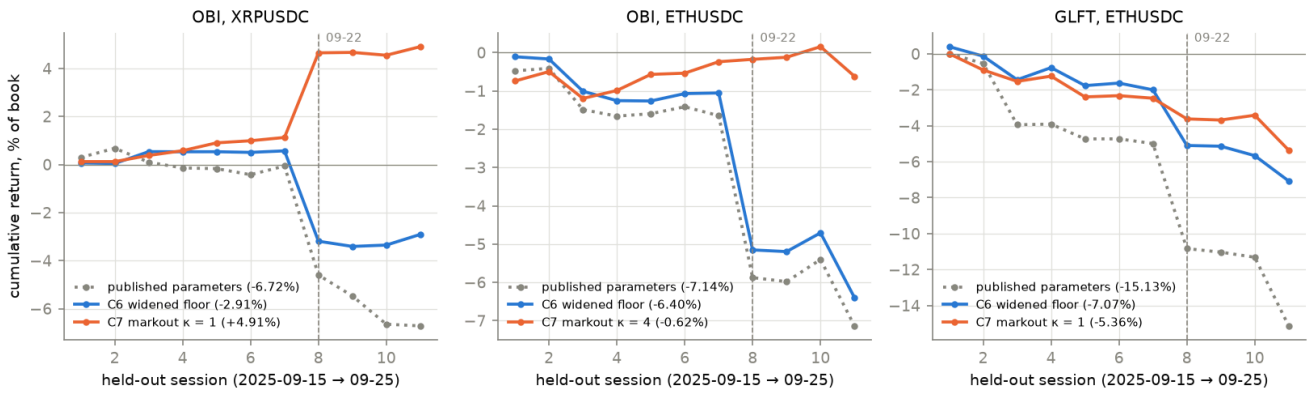


Figure — cumulative return over the eleven held-out sessions: the published parameters (dotted), C6's widened spread floor (blue), and C7's markout rule on top of it (orange). Session 8 is 2025-09-22, when the market crashed.

Two things are visible at once. The widened floor helps on every panel but does not stop the crash-day loss. The markout rule does: on 09-22 the widened strategies lose 3–4% while the markout versions lose about 1% or gain — plausibly because they had already widened as their fills turned toxic. Part of XRP's gain that day is holding the right inventory through the move, which is why the result is also quoted with 09-22 removed.

3. Every method, held out

asset	strategy	published	C6 widened	C6 RL v2	C7 markout	C4 grid (20 d)	C4 GRPO
ETH	glft	−15.13	−7.07	−6.30	−5.36	−3.01	**−1.89**
ETH	hfgrid	−1.98	−2.11	−2.20	−1.99	**−1.86**	–
ETH	obi	−7.14	−6.40	−8.00	**−0.62**	**−0.62**	−2.91
XRP	glft	−13.17	−1.89	−0.86	**−0.55**	−1.22	–
XRP	hfgrid	−1.58	−1.15	−1.48	**+1.96**	+0.59 †	–
XRP	obi	−6.72	−2.91	−2.15	**+4.91**	+3.64	–
XRP	apt	−3.39	+2.00	−1.66	–	+7.26 †	–
XRP	basis	−4.59	+0.36	−2.35	–	+1.69 †	–

Held-out cumulative return, % of book, eleven sessions. RL and GRPO are medians over seeds. † trades a handful of fills or none on a median session; the number is one crash session, not market making.

3.1 On C8's split, with stress tests

C8 moved to 24 training sessions and kept the last seven held out, then stress-tested everything that made money: maker fee without the rebate, higher latency, pessimistic queue models, neighbouring parameters, and without the crash or the best session.

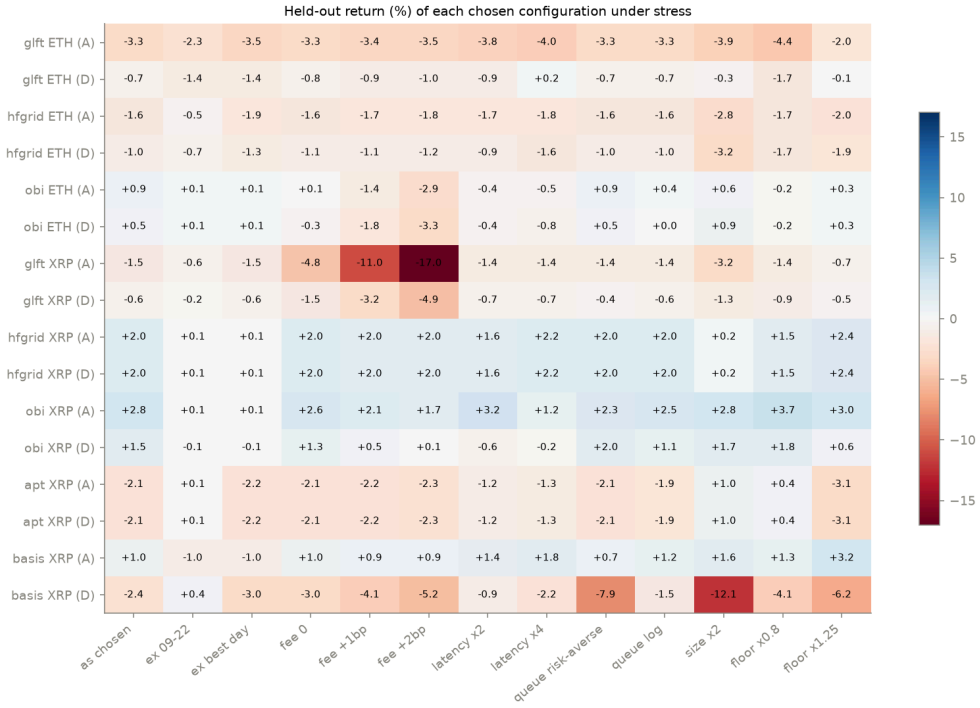


Figure — from C8 §6: held-out return of each chosen configuration under each stress.

configuration	held out	ex 09-22	ex best day	fee 0	fee +2 bp	latency ×4	fills/day	verdict
SUIUSDC OBI (floor 8 bp, κ 4)	**+2.52**	**+1.77**	**+1.52**	**+2.05**	+0.21	**+1.30**	175	makes money on ordinary sessions, robust to costs; sensitive to κ
XRPUSDC OBI (floor 12 bp, κ 2, skew ×2)	+2.84	+0.09	+0.09	+2.61	+1.68	+1.17	106	robust to costs; the gain is the crash
XRPUSDT OBI (floor 8 bp, κ 2)	+3.53	+0.48	+0.48	+3.17	+1.74	+2.84	189	as XRPUSDC
AVAXUSDC OBI (floor 16 bp, κ 0)	+2.68	+2.34	−0.37	+2.35	+1.04	+3.14	150	one session
ETHUSDC OBI (floor 6 bp, κ 4)	+0.85	+0.14	+0.14	+0.09	−2.92	−0.48	369	the rebate
BNBUSDC GLFT (floor 2 bp, κ 1)	+1.72	+1.53	+1.23	+0.75	−3.05	+1.71	489	the rebate at higher fees

Held-out cumulative return, % of book, seven sessions (09-19 to 09-25).

4. What each phase contributed

phase	question	answer, held out
C1	Does calibrated GLFT make money here?	No: it quotes inside the cost of adverse selection and loses 0.8 bp per fill.
C2	Can the problem be posed for RL without a degenerate answer?	Yes, once the "quote so far out nothing fills" corner is ruled out.
C3	Do DDPG/TD3/SAC beat the fixed formula?	They lose less than it, never make money; seed spread is large.
C6	Is that just GLFT being a bad baseline?	No. On five strategies and nine assets, a per-asset wider spread is the largest single gain (+2.25 points median); TD3 on top of it does not beat it.
C7	What information is the policy missing?	Not the deep book or BTCUSDT. The toxicity of its own recent fills: a markout rule beats C6 in 6 of 6 pairs and 13 of 18 untuned.
C4	Can search (grid, GRPO, PPO) tune the fixed rules further?	The grid refines them on ETH; learned search, fill targets, an inventory-discounted reward and larger networks do not.
C9	Can the fair price be forecast ~1 s ahead, and does denoising help anywhere?	Foresight of 1–3 s would be worth up to +40 points held out (little on XRP); real forecasts reach R^2 0.02–0.10 and help one asset (AVAX). DAgger with a future-seeing teacher halves ETH GLFT's loss but does not make it pay. No denoiser helps on any of nine assets.

phase	question	answer, held out
C8	Do risk controls, denoising or RL on top of the markout rule help, and does what makes money survive stress?	A circuit breaker helps the losers on ETH; no denoiser, combination of controls or RL policy improves OBI. SUIUSDC OBI survives every cost, latency and session stress but κ ×0.5; XRP's gain is the crash, ETH's the rebate.

5. Lessons that held up

1. **Measure the cost of a fill before tuning anything.** Every positive result in Part C comes from quoting wider when fills are expensive.
2. **Fixed rules first.** Each learned method was beaten by a hand-written rule that encoded the right quantity; RL is worth trying on top of the best rule, not instead of it.
3. **Seeds and sessions, not single runs.** Single-seed RL rows were systematically kinder than five-seed rows, and four training sessions chose XRP parameters that twenty did not.
4. **Report the crash day separately.** One session (2025-09-22) can make a strategy that barely trades look like the best in the table.
5. **Fills are not free.** Both forcing volume and rewarding it cost money on this venue, at every level tried.
6. **Stress-test before believing a gain.** Of six configurations that made money held out in C8, one survives removing the crash, the best session and the rebate; the others were one session or the rebate.
7. **More knobs fit the training month.** Adding controls chosen on 24 sessions helped strategies that lose and hurt the one that works.
8. **Measure the ceiling before learning.** C9's oracle showed where the money is (the next 1–3 s) and its noisy oracles how good a forecast must be ($R^2 \approx 0.4$); real forecasts at 0.02–0.10 were then no surprise.

6. RL in Part C: what was tried, why it failed, what could make it tractable

6.1 What was tried

method (phase)	learns from	the policy sets	decides every	held out
DDPG, TD3, SAC (C3)	reward: mark-to-market PnL minus an inventory penalty	GLFT's γ, c_1 and a spread scale	quoting step	loses less than GLFT, never makes money
TD3 residual policy, v1/v2 (C6)	reward relative to the rule it adjusts (counterfactual in v2)	multipliers on the rule's spread, skew, signal and grid	5 s	never beats the widened rule
PPO- and GRPO-style search (C4)	ranked session PnL of sampled parameter sets	the rule's parameters for a whole session	session	the plain grid refines ETH; learned search does not

method (phase)	learns from	the policy sets	decides every	held out
fill targets, edge/inventory reward, networks up to 170× (C4)	the same, reshaped	the same	5 s	no change
TD3 on top of the markout rule (C8)	reward relative to the rule, toxicity in the state	the same multipliers	5 s	never beats the rule
DAgger with a privileged teacher (C9)	a teacher's label at every quote (the teacher sees the future mid)	pull or widen one side	100 ms	halves ETH GLFT's loss, does not help ETH OBI

6.2 Why RL kept failing, and what DAgger changes

- **Reward-driven RL learns from a signal buried in noise.** One quoting decision is worth a fraction of a basis point; the position's mark-to-market moves by basis points a second. The learner has to credit a gain to one of thousands of decisions inside that noise, which is why seed spreads were 1–10 points and why larger networks changed nothing (the network was never the bottleneck).
- **DAgger replaces the reward with a teacher's label at every step.** It turns the problem into supervised classification with dense, low-variance feedback, and relabels the states the student itself visits (the "DAgger" part, which fixes plain imitation's drift into unfamiliar states). It removes the noise problem completely.
- **No method can beat the information in its inputs.** A student can learn at best the teacher's action averaged over every situation that looks the same from its observations; when the teacher's decision depends on the future and the book barely hints at it, that average is weak. C9 measured the limit: the future mid is worth up to +40 points, a forecast needs $R^2 \approx 0.4$ to pay, and the book supplies 0.02–0.10. RL sees the same inputs with a noisier signal, so it hits the same limit or a lower one.

6.3 Clues that would make the search tractable

Each attacks the noise and credit-assignment problem; none adds information.

1. **A privileged critic (asymmetric actor-critic).** In TD3 or SAC the critic is used only in training, so it may see the future mid, the teacher's labels and each fill's eventual markout; the actor sees only what the quoter sees. This cuts the variance of the learning signal directly. The first thing to try.
2. **Imitation first, RL second.** Initialise from the rule or the DAgger student and fine-tune with a penalty on drifting away from it (the recipe that makes RL work in robotics and games), instead of exploring from scratch — which in market making means paying the spread to learn.
3. **Imitate values, not actions (AggreVaTe).** The teacher gives each action's cost-to-go instead of its choice, which fixes what broke C9's first teacher (a 5 s markout instead of what a fill actually earns). It needs counterfactual replays per labelled state, so it would run on a sample of a few thousand states per session.

4. **A curriculum over privileged information.** Train with a noisy oracle of the future move as an input and lower its R^2 towards what real forecasts reach, so the policy first learns what to do with good information and then how to discount weak information. C9's noisy oracle is exactly this input.
5. **More shaping and variance reduction.** Beyond the counterfactual reward (C6) and the edge/inventory split (C4): rewards built on markout curves, and every comparison paired on the same sessions.
6. **Pool assets.** One policy over nine assets sees nine times the rare large moves, which a single asset starves.

6.4 Are the strategy parameters too rigid?

- **The RL action spaces were, for what C9 found valuable.** The oracle's gain comes from pulling one side of the book 1–3 s before a move. The RL policies decided every 5 s (too slow), scaled both sides at once (could not pull one), and stayed inside the spread floor. C9's kernel inputs — a series read every 100 ms, per-side widening and pulling — are the missing actions, and no RL policy has used them yet.
- **But every added degree of freedom has so far fit the training month.** The greedy combination of controls (C8), bigger networks (C4), the denoisers and forecast fair prices on SUIUSDC, XRPUSDT and BNB (C9) all gained on training sessions and lost held out. With 24 training sessions the rigid parametric rules act as the regularisation.
- **And rigidity is not what blocks the gain.** The same kernel reaches +6.53% on ETH OBI when told the future, and the DAgger student had exactly the right actions. The strategy can express the profitable policy; the inputs do not say when to use it.

Next RL experiment, if Part C continues: TD3 with a privileged critic, C9's per-side 100 ms actions, initialised from the rule, pooled across assets, five seeds, under the usual protocol. The honest expectation is roughly the DAgger result (ETH GLFT's loss halved, OBI unchanged) unless the inputs change — which is §7.

7. Data from outside the order book

C9 found that the value is in rare large moves (the top 1% of 3 s moves) that the traded contract's own book does not anticipate. Large moves usually start somewhere else: on the more liquid contract for the same asset, on spot, on another venue, or in a liquidation cascade. What we would want, ranked by expected value at a 1–3 s horizon, and what can be had for free (availability checked on 2026-09-30 for the study month, 2025-08-26 to 09-25):

data	why it could lead	free source	resolution	checked
The same asset's USDT perpetual (ETHUSDT, XRPUSDT, SUIUSDT … for our USDC contracts)	price discovery happens on the far larger USDT contract; the USDC contract follows	Binance public archive, `data.binance.vision/data/futures/um/daily/trades` (and `aggTrades`)	every trade, ms	✓ available for the month and through 2026-09; ETHUSDT 34 MB/day zipped

data	why it could lead	free source	resolution	checked
Spot (Binance ETHUSDT, ETHUSDC, XRPUSDT …)	the index and mark price are built from spot; spot leads some moves	Binance archive, `data/spot/daily/trades`	every trade, ms	✓ ETHUSDT spot 18 MB/day
Other venues' perpetuals and spot (Bybit, OKX)	large moves often start on one venue and arbitrage carries them over	Bybit `public.bybit.com/trading/` and `/spot/`; OKX `okx.com/cdn/okex/traderecords/trades/daily/`	every trade (Bybit 0.1 ms)	✓ Bybit ETHUSDT 46 MB/day; ✓ OKX files served
Coinbase spot (ETH-USD, XRP-USD)	the main US venue, leads in US hours	Coinbase Exchange REST `/products/{id}/trades` (free, paginated, slow)	every trade	not checked
Liquidations	cascades drive the largest moves	no free history: Binance's archive has none, Tardis's free samples are blocked here (below); **free live**: Binance `!forceOrder@arr` (at most one per second per symbol), Bybit `allLiquidation`, OKX liquidation orders	event	live only
Funding, premium, mark and index price	funding-time flows; basis stress	Binance archive: `monthly/fundingRate`, daily `premiumIndexKlines`, `markPriceKlines`, `indexPriceKlines`	8 h; 1 min	✓
Open interest, long/short and taker ratios	positioning regime	Binance archive, daily `metrics`	5 min	✓
Depth ±1–5%	book resilience regime	Binance archive, daily `bookDepth`	~30 s	✓
Implied volatility (Deribit DVOL, options)	volatility regime	Deribit public API	1 min and slower	not checked
Best bid/ask of other contracts	quote-level lead-lag	Binance `bookTicker` archive	tick	✗ discontinued after 2024-03-30
Full L2 books of other contracts or venues	book-level lead-lag	Tardis free files (first day of each month, no key): one request succeeded, then Cloudflare returned 403 to every request from this machine; **free**: record live websockets with hftbacktest's collector	tick	partial

data	why it could lead	free source	resolution	checked
Our own L2 after 2025-09-25	the forward test (a backtest needs the book, not only trades)	the project's dataset (`cardinalai/backtest_data` on Hugging Face, needs `HF_TOKEN` to list); or record live	tick	not checked

Only the first rows can trigger a 1–3 s decision; the slow series (funding, open interest, depth snapshots, implied volatility) can only condition the regime — when to be wider — which the markout rule and the circuit breaker already do from our own flow.

How it would be tested, with the C9 machinery unchanged:

1. **Alignment first.** The archives carry exchange timestamps; the backtest sees our own book at local receive time. External events are shifted by an assumed feed latency (tens of ms same-venue, 50–150 ms cross-venue) before they may enter a feature, and every result is re-run with +100 ms and +250 ms extra delay as a stress: a lead that disappears under 250 ms of delay is not tradable.
2. **Offline, cheap:** add lead-lag features (returns and signed flow of the USDT perp, spot and Bybit over the last 0.1–1 s; the basis between them; bursts of large trades) to `predict.py`, and measure held-out R^2 and, more to the point, the AUC and precision of a top-1% large-move classifier against the book-only 0.71 and 3–11%.
3. **Only if (2) moves:** the large-move rule and the DAgger student with the new inputs, chosen on training sessions, held out, stress-tested.
4. **Disk:** a month of one source is 0.5–1.5 GB zipped; reduced on the fly to the 100 ms grid it is a few MB, so the raw files are downloaded, reduced and deleted day by day.

8. What next

C8 did the follow-ups listed before it; C9 answered the fair-price and denoising questions. What remains in Part C, cheapest and most informative first:

- **The forward test:** the frozen SUIUSDC and XRP OBI choices (and AVAXUSDC with C9's boosted-tree fair price) on sessions after 2025-09-25, with no re-selection. Seven held-out sessions cannot tell a +2.5% edge from luck. Needs the book after 09-25 (§7, last row).
- **Lead-lag inputs from outside the book** (§7): the only line of work that could raise the information limit every method has hit.
- **RL with a privileged critic and C9's per-side actions** (§6.4): worth running only once the inputs carry more than R^2 0.1.
- **Smaller:** OBI on more assets; κ from the asset's own markout (SUIUSDC depends on $\kappa = 4$ and only the grid chooses it); a crash-aware breaker.

Everything on the original ladder's Part C list is done. Part C can pause here without losing anything: the cache, the features and the kernel inputs are all on the branch, and each item above starts from them.